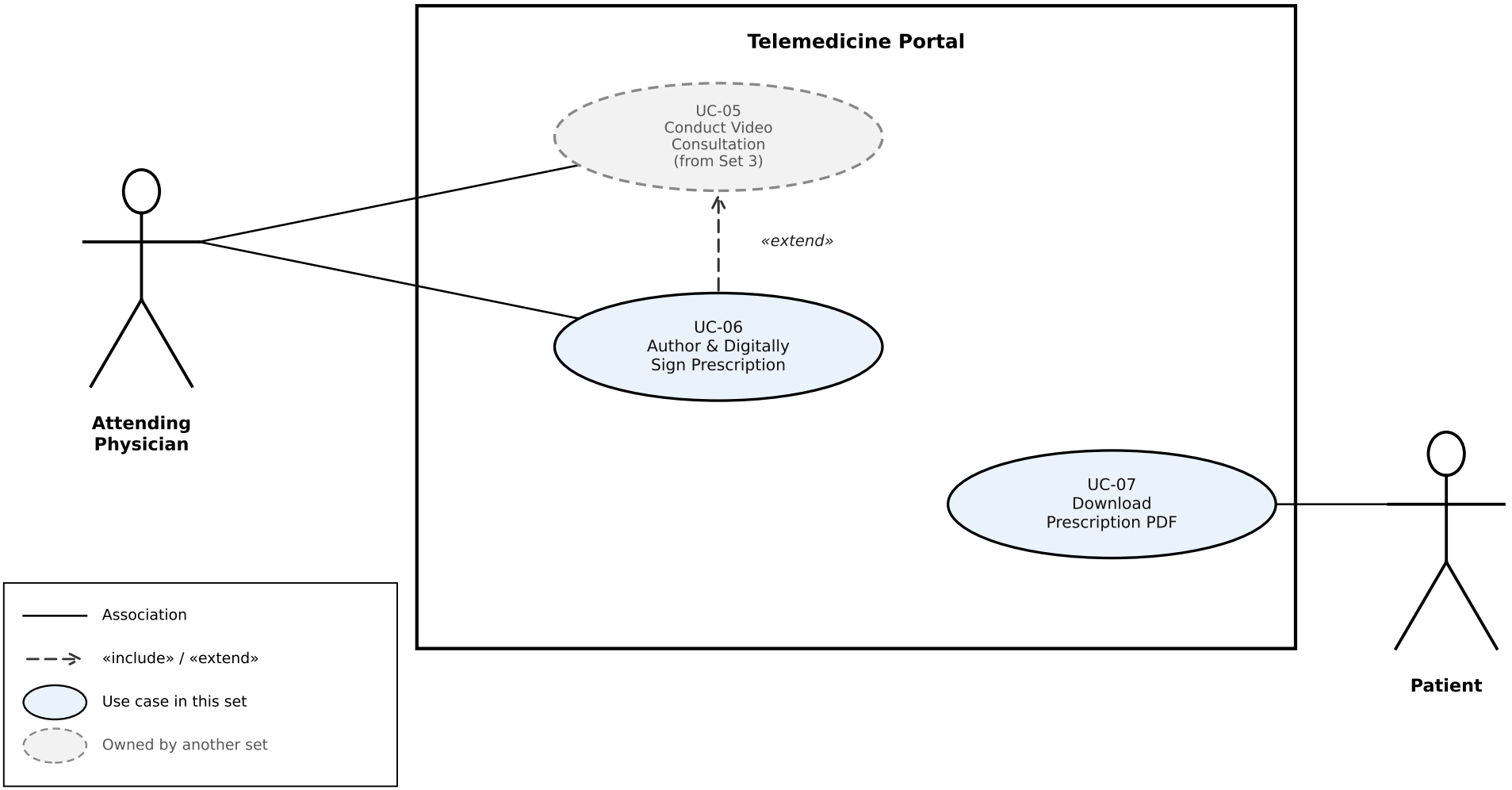
Set 4 — Prescription Lifecycle

Weightage: 25% | Use cases: UC-06, UC-07



Use-Case Flow Specification

Telemedicine Slot Booking & Prescription Portal

Set 4 — Prescription Lifecycle | Weightage: 25%

Use Case: UC-06 — Author & Digitally Sign Prescription *(extends UC-05: Conduct Video Consultation, owned by Set 3)*

Primary Actor

Attending Physician

Preconditions

- A consultation for this patient has been completed and marked complete by the system.
- The physician is authenticated and holds a valid, unexpired medical licence on file.
- The physician's digital signing certificate is active and not revoked.

Postconditions

- Success: A cryptographically signed prescription PDF is stored against the patient record and released for download (UC-07).
- Failure: No prescription is issued; the draft is retained and the failure reason is written to the audit log.

Main Success Scenario

1. Physician opens the completed consultation record and selects “Create Prescription”.
2. System displays a structured prescription form pre-filled with patient and consultation details.
3. Physician enters each drug with dosage, frequency, duration, and any special instructions.
4. System validates the entries against the drug formulary and flags dosage or interaction warnings.
5. Physician reviews the warnings, adjusts the entries if required, and submits the prescription.
6. System re-validates the physician's licence status and signing certificate.
7. System applies the physician's digital signature and compiles the prescription into a PDF.
8. System stores the signed PDF against the patient record, writes an audit entry, and notifies the patient that it is available for download.
9. Use case ends successfully.

Alternate Flow — 6a. Licence or Signing Certificate Invalid

a. 6a1. System finds the physician's licence expired or the signing certificate revoked.

b. 6a2. System blocks signing, saves the prescription as an unsigned draft, and displays the specific reason to the physician.

c. 6a3. System writes the blocked attempt to the audit log and alerts the clinic administrator; no PDF is generated and the patient is not notified.